

Transformer Oil Analysis

Water Content & DGA Diagnostic Report

Generated: 16-06-2026 00:31 | Method: Karl Fischer (IS 13567 / ASTM D1533 / IEC 60814) | Limits: TRU-FIL 40 ppm default, SGS/CPRI & Generic 45 ppm default, PDF-stated limit always takes priority | No AI / external API used.

Total Reports	Unacceptable	Acceptable	Not Detected
11	5	6	0

Report Index

#	Equipment	Format	Water (ppm)	Verdict
1	ODYSSEY S/s / TR#1	TRU-FIL	72.0	UNACCEPTABLE
2	GLEN HEIGHT / TR#2	TRU-FIL	80.0	UNACCEPTABLE
3	GLEN HEIGHT / TR#2	TRU-FIL	67.0	UNACCEPTABLE
4	ODYSSEY S/s / TR#1	TRU-FIL	89.0	UNACCEPTABLE
5	MARWAHA CENTRAL S/s / TR#2	TRU-FIL	91.0	UNACCEPTABLE
6	ZENTA S/s / TR#2	TRU-FIL	28.0	ACCEPTABLE
7	AVLON S/s / TR-1	TRU-FIL	22.0	ACCEPTABLE
8	ODYSSEY S/s / TR#2	TRU-FIL	23.0	ACCEPTABLE
9	ODYSSEY S/s / TR#2	TRU-FIL	30.0	ACCEPTABLE
10	VERONA S/s / TR#1	TRU-FIL	11.0	ACCEPTABLE
11	S.L.M / TR#2	TRU-FIL	30.0	ACCEPTABLE

1. ODYSSEY S/s / TR#1

[UNACCEPTABLE] Equipment-Main Tank - Bottom Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02684-TataPower-East_Suburb-ODYSSEY_S_s-TR#1-Equipment-Main_Tank_-_Bottom.pdf

[UNACCEPTABLE] Unacceptable - exceeds 50 ppm limit. Corrective action required.

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	72.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	50 ppm Max	PDF-stated limit (TRU-FIL format row) - 50 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Immediate Actions Required

(1) Hot-oil circulation / vacuum dehydration / Fuller's Earth filtration — target well below 50 ppm post-treatment. (2) Inspect breather, gaskets and conservator seal for moisture ingress. (3) Verify BDV and DDF before continued operation at rated load. (4) Re-test after filtration and trend the result; if moisture persists after two passes, assess paper-insulation moisture and consider oil replacement.

Dissolved Gas Analysis (DGA)

Gas	Value (ppm v/v)	Reference Limit	Status
Carbon Monoxide (CO)	182.97	400	OK

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.01 mN/m	0.02 mN/m	OUT OF RANGE
Dielectric Dissipation Factor (DDF, 90C)	0.24 %	0.50 %	OK
Acidity	0.52 mg KOH/g	50.00 mg KOH/g	OK

Equipment & Report Details

Equipment	ODYSSEY S/s / TR#1	Equipment Type	-
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02684	Report Date	30-05-2026
Sampling Pt.	Equipment-Main Tank - Bottom Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02684-TataPower-E

2. GLEN HEIGHT / TR#2

[UNACCEPTABLE] Equipment-Main Tank - Bottom Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02687-TataPower-East_Suburb-GLEN_HEIGHT-TR#2-Equipment-Main_Tank_-_Bottom.pdf

[UNACCEPTABLE] Unacceptable - exceeds 50 ppm limit. Corrective action required.

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	80.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	50 ppm Max	PDF-stated limit (TRU-FIL format row) - 50 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Immediate Actions Required

(1) Hot-oil circulation / vacuum dehydration / Fuller's Earth filtration — target well below 50 ppm post-treatment. (2) Inspect breather, gaskets and conservator seal for moisture ingress. (3) Verify BDV and DDF before continued operation at rated load. (4) Re-test after filtration and trend the result; if moisture persists after two passes, assess paper-insulation moisture and consider oil replacement.

Dissolved Gas Analysis (DGA)

Gas	Value (ppm v/v)	Reference Limit	Status
Carbon Monoxide (CO)	284.20	400	WATCH

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.01 mN/m	0.02 mN/m	OUT OF RANGE
Dielectric Dissipation Factor (DDF, 90C)	0.16 %	0.50 %	OK
Acidity	6.85 mg KOH/g	50.00 mg KOH/g	OK

Equipment & Report Details

Equipment	GLEN HEIGHT / TR#2	Equipment Type	-
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02687	Report Date	04-06-2026
Sampling Pt.	Equipment-Main Tank - Bottom Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02687-TataPower-E

3. GLEN HEIGHT / TR#2

[UNACCEPTABLE] Equipment-OLTC Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02688-TataPower-East_Suburb-GLEN_HEIGHT-TR#2-Equipment-OLTC.pdf

[UNACCEPTABLE] Unacceptable - exceeds 0 ppm limit. Corrective action required.

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	67.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	0 ppm Max	PDF-stated limit (TRU-FIL format row) - 0 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Immediate Actions Required

(1) Hot-oil circulation / vacuum dehydration / Fuller's Earth filtration — target well below 0 ppm post-treatment. (2) Inspect breather, gaskets and conservator seal for moisture ingress. (3) Verify BDV and DDF before continued operation at rated load. (4) Re-test after filtration and trend the result; if moisture persists after two passes, assess paper-insulation moisture and consider oil replacement.

Dissolved Gas Analysis (DGA)

No DGA gas values were detected in this PDF.

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.02 mN/m	0.02 mN/m	OUT OF RANGE
Dielectric Dissipation Factor (DDF, 90C)	0.13 %	0.50 %	OK
Acidity	—	—	—

Equipment & Report Details

Equipment	GLEN HEIGHT / TR#2	Equipment Type	OLTC
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02688	Report Date	30-05-2026
Sampling Pt.	Equipment-OLTC Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02688-TataPower-E

4. ODYSSEY S/s / TR#1

[UNACCEPTABLE] Equipment-OLTC Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02689-TataPower-East_Suburb-ODYSSEY_S_s-TR#1-Equipment-OLTC.pdf

[UNACCEPTABLE] Unacceptable - exceeds 0 ppm limit. Corrective action required.

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	89.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	0 ppm Max	PDF-stated limit (TRU-FIL format row) - 0 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Immediate Actions Required

(1) Hot-oil circulation / vacuum dehydration / Fuller's Earth filtration — target well below 0 ppm post-treatment. (2) Inspect breather, gaskets and conservator seal for moisture ingress. (3) Verify BDV and DDF before continued operation at rated load. (4) Re-test after filtration and trend the result; if moisture persists after two passes, assess paper-insulation moisture and consider oil replacement.

Dissolved Gas Analysis (DGA)

No DGA gas values were detected in this PDF.

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.02 mN/m	0.02 mN/m	OUT OF RANGE
Dielectric Dissipation Factor (DDF, 90C)	0.13 %	0.50 %	OK
Acidity	—	—	—

Equipment & Report Details

Equipment	ODYSSEY S/s / TR#1	Equipment Type	OLTC
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02689	Report Date	30-05-2026
Sampling Pt.	Equipment-OLTC Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02689-TataPower-E

5. MARWAHA CENTRAL S/s / TR#2

[UNACCEPTABLE] Equipment-OLTC Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02691-TataPower-East_Suburb-MARWAHA_CENTRAL_S_s-TR#2-Equipment-OLTC.pdf

[UNACCEPTABLE] Unacceptable - exceeds 40 ppm limit. Corrective action required.

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	91.0 ppm	Extractor: trufil_primary
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	40 ppm Max	IS 1866:2017 Table-5 default (<220 kV, TRU-FIL format) - 40 ppm
Source (Vendor) Verdict	-	As stated in PDF (if present)

Immediate Actions Required

(1) Hot-oil circulation / vacuum dehydration / Fuller's Earth filtration — target well below 40 ppm post-treatment. (2) Inspect breather, gaskets and conservator seal for moisture ingress. (3) Verify BDV and DDF before continued operation at rated load. (4) Re-test after filtration and trend the result; if moisture persists after two passes, assess paper-insulation moisture and consider oil replacement.

Dissolved Gas Analysis (DGA)

No DGA gas values were detected in this PDF.

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.01 mN/m	0.02 mN/m	OUT OF RANGE
Dielectric Dissipation Factor (DDF, 90C)	0.02 %	0.50 %	OK
Acidity	—	—	—

Equipment & Report Details

Equipment	MARWAHA CENTRAL S/s / TR#2	Equipment Type	OLTC
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02691	Report Date	04-06-2026
Sampling Pt.	Equipment-OLTC Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02691-TataPower-E

6. ZENTA S/s / TR#2

[ACCEPTABLE] Equipment-OLTC Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02681-TataPower-East_Suburb-ZENTA_S_s-TR#2-Equipment-OLTC.pdf

[ACCEPTABLE] Acceptable - <= 40 ppm limit

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	28.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	40 ppm Max	PDF-stated limit (TRU-FIL format row) - 40 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Dissolved Gas Analysis (DGA)

Gas	Value (ppm v/v)	Reference Limit	Status
Carbon Monoxide (CO)	73.63	400	OK

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.03 mN/m	0.02 mN/m	OK
Dielectric Dissipation Factor (DDF, 90C)	0.02 %	0.50 %	OK
Acidity	1.09 mg KOH/g	50.00 mg KOH/g	OK

Equipment & Report Details

Equipment	ZENTA S/s / TR#2	Equipment Type	OLTC
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02681	Report Date	27-05-2026
Sampling Pt.	Equipment-OLTC Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02681-TataPower-E

7. AVLON S/s / TR-1

[ACCEPTABLE] Equipment-OLTC Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02682-TataPower-East_Suburb-AVLON_S_s-TR-1-Equipment-OLTC.pdf

[ACCEPTABLE] Acceptable - <= 40 ppm limit

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	22.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	40 ppm Max	PDF-stated limit (TRU-FIL format row) - 40 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Dissolved Gas Analysis (DGA)

No DGA gas values were detected in this PDF.

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.02 mN/m	0.02 mN/m	OK
Dielectric Dissipation Factor (DDF, 90C)	0.02 %	0.50 %	OK
Acidity	—	—	—

Equipment & Report Details

Equipment	AVLON S/s / TR-1	Equipment Type	OLTC
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02682	Report Date	27-05-2026
Sampling Pt.	Equipment-OLTC Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02682-TataPower-E

8. ODYSSEY S/s / TR#2

[ACCEPTABLE] Equipment-Main Tank - Bottom Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02685-TataPower-East_Suburb-ODYSSEY_S_s-TR#2-Equipment-Main_Tank_-_Bottom.pdf

[ACCEPTABLE] Acceptable - <= 40 ppm limit

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	23.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	40 ppm Max	PDF-stated limit (TRU-FIL format row) - 40 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Dissolved Gas Analysis (DGA)

Gas	Value (ppm v/v)	Reference Limit	Status
Carbon Monoxide (CO)	83.14	400	OK

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.02 mN/m	0.02 mN/m	OK
Dielectric Dissipation Factor (DDF, 90C)	0.01 %	0.50 %	OK
Acidity	0.71 mg KOH/g	50.00 mg KOH/g	OK

Equipment & Report Details

Equipment	ODYSSEY S/s / TR#2	Equipment Type	-
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02685	Report Date	30-05-2026
Sampling Pt.	Equipment-Main Tank - Bottom Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02685-TataPower-E

9. ODYSSEY S/s / TR#2

[ACCEPTABLE] Equipment-OLTC Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02686-TataPower-East_Suburb-ODYSSEY_S_s-TR#2-Equipment-OLTC.pdf

[ACCEPTABLE] Acceptable - <= 40 ppm limit

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	30.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	40 ppm Max	PDF-stated limit (TRU-FIL format row) - 40 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Dissolved Gas Analysis (DGA)

No DGA gas values were detected in this PDF.

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.02 mN/m	0.02 mN/m	OK
Dielectric Dissipation Factor (DDF, 90C)	0.01 %	0.50 %	OK
Acidity	—	—	—

Equipment & Report Details

Equipment	ODYSSEY S/s / TR#2	Equipment Type	OLTC
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02686	Report Date	30-05-2026
Sampling Pt.	Equipment-OLTC Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02686-TataPower-E

10. VERONA S/s / TR#1

[ACCEPTABLE] Equipment-Main Tank - Bottom Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02690-TataPower-East_Suburb-VERONA_S_s-TR#1-Equipment-Main_Tank_-_Bottom.pdf

[ACCEPTABLE] Acceptable - <= 40 ppm limit

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	11.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	40 ppm Max	PDF-stated limit (TRU-FIL format row) - 40 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Dissolved Gas Analysis (DGA)

Gas	Value (ppm v/v)	Reference Limit	Status
Carbon Monoxide (CO)	75.12	400	OK

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.03 mN/m	0.02 mN/m	OK
Dielectric Dissipation Factor (DDF, 90C)	0.00 %	0.50 %	OK
Acidity	0.52 mg KOH/g	50.00 mg KOH/g	OK

Equipment & Report Details

Equipment	VERONA S/s / TR#1	Equipment Type	-
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02690	Report Date	27-05-2026
Sampling Pt.	Equipment-Main Tank - Bottom Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02690-TataPower-E

11. S.L.M / TR#2

[ACCEPTABLE] Equipment-OLTC Rating 1250 KVA | Copy of Oil-Test-Reports-TL-OTR-26-02693-TataPower-East_Suburb-S.L.M-TR#2-Equipment-OLTC.pdf

[ACCEPTABLE] Acceptable - <= 40 ppm limit

Parameter	Value	Basis / Notes
Water Content (Karl Fischer)	30.0 ppm	Extractor: trufil_full_row
Detected Format	TRU-FIL	TRU-FIL / SGS-CPRI / Generic
Applicable Limit	40 ppm Max	PDF-stated limit (TRU-FIL format row) - 40 ppm Max
Source (Vendor) Verdict	Acceptable	As stated in PDF (if present)

Dissolved Gas Analysis (DGA)

No DGA gas values were detected in this PDF.

Physical & Chemical Tests

Test	Value	Limit	Status
Breakdown Voltage (BDV)	—	—	—
Interfacial Tension (IFT)	0.03 mN/m	0.02 mN/m	OK
Dielectric Dissipation Factor (DDF, 90C)	0.04 %	0.50 %	OK
Acidity	—	—	—

Equipment & Report Details

Equipment	S.L.M / TR#2	Equipment Type	OLTC
Voltage Class	24 kV	Rating	1250 KVA
Report No.	TL/OTR/26/02693	Report Date	02-06-2026
Sampling Pt.	Equipment-OLTC Rating 1250 KVA	Insul. Fluid	Dielectric Fluid-Mineral Base
Limit Standard	IS 1866	Owner	Tata Power Company Limited
Location	, Near MTNL Hiranandani , Swami Narayan	Source File	Copy of Oil-Test-Reports-TL-OTR-26-02693-TataPower-E